

Analytical problem solving Activity

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Course: Computer vision

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1. Qm:

$$C_x = 24, \quad C_y = 18$$

Gradient Magnitude:

$$\begin{aligned} G &= \sqrt{C_x^2 + C_y^2} \\ &= \sqrt{24^2 + 18^2} \\ &= \sqrt{576 + 324} = \sqrt{900} = 30 \end{aligned}$$

Gradient magnitude = 30

Gradient Direction:

$$\theta = \tan^{-1} \left(\frac{C_y}{C_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{18}{24} \right)$$

$$\theta = \tan^{-1} (0.75) \approx 36.87.$$

Gradient Direction = 36.87.

2. Linear Gradient magnitude.

Pixel	P_1	P_2	P_3	P_4	P_5
magnitude	25	50	90	60	30

since the gradient direction is horizontal each pixel is compared with its left and right.

$P_1 = 25 \rightarrow$ boundary pixel $\rightarrow$ retained

$P_2 = 50 \rightarrow$ less than $P_3 (90) \rightarrow$ suppressed.

$P_3 = 90 \rightarrow$ Greater than $P_2 (50)$ and $P_4 (60) \rightarrow$ retained

$P_4 = 60 \rightarrow$ less than $P_3 (90) \rightarrow$ suppressed.

$P_5 = 30 \rightarrow$ Boundary pixel $\rightarrow$ retained.

Result After NMS:

Pixel	P_1	P_2	P_3	P_4	P_5
NMS result	25	0	90	0	30

Remaining edge pixels: P_1 , P_3 and P_5 .

3. Double Thresholding.

Ans:

high threshold = 100

low threshold = 50

Rule:

magnitude $\geq 100 \rightarrow$ strong edge

$50 \leq \text{magnitude} < 100 \rightarrow$ weak edge.

Pixel	magnitude	classification
P_1	135	strong edge
P_2	90	weak edge
P_3	45	non-edge
P_4	120	strong edge
P_5	70	weak edge
P_6	20	non-edge.

Answer:

strong edge : P_1, P_4 .

weak edge : P_2, P_5

non-edge : P_3, P_6 .

4

Pixel	P_1	P_2	P_3	P_4	P_5	P_6	P_7
magnitude	130	95	110	115	65	25	10.

thresholds.

high = 100

low = 50.

1. strong edge.

magnitude ≥ 100 : P_1, P_4, P_9 .

2. weak edge.

$50 \leq \text{magnitude} < 100$: P_2, P_5 .

3. non-edges.

magnitude < 50 : P_3, P_6 .

4. final edge pixels after hysteresis.

P_2 is connected to strong edge P_1 , P_2 is retained.

P_5 is not connected to any strong edge so P_5 is removed.

final edges : P_1, P_2, P_4, P_9 .

5. Hough Circle Transform:

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

i) edge points.

$$P_1 = (5, 4) \quad , \quad P_2 = (7, 6)$$

$$C_1 = (2, 4) \quad , \quad C_2 = (4, 3)$$

for $C_1 = (2, 4)$

$$\begin{aligned} P_1 \text{ and } C_1 : r &= \sqrt{(5-2)^2 + (4-4)^2} \\ &= \sqrt{3^2 + 0^2} = \sqrt{9} = 3. \end{aligned}$$

$$\boxed{r=3}$$

$$\begin{aligned} P_2 \text{ and } C_2 : r &= \sqrt{(7-2)^2 + (6-4)^2} \\ &= \sqrt{25+4} = \sqrt{29} = 5.39 \end{aligned}$$

$$\boxed{r=5.39}$$

$$\begin{aligned} P_1 \text{ and } C_2 : r &= \sqrt{(5-4)^2 + (4-3)^2} \\ &= \sqrt{1+1} = \sqrt{2} = 1.41 \end{aligned}$$

$$r = 1.41$$

$$P_2 \text{ and } C_2 : r = \sqrt{(7-4)^2 + (6-3)^2}$$

$$= \sqrt{9+9} = \sqrt{18} = 4.24$$

$$\boxed{r = 4.24}$$

6. Calculate the radius:

Ans:

$$r = (8, 6) \quad , \quad C = (4, 3)$$

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

$$= \sqrt{(8-4)^2 + (6-3)^2}$$

$$= \sqrt{25}$$

$$\boxed{r = 5}$$

7. Calculate the Radius.

$$C = (5, 4) \quad P = (8, 8)$$

$$r = \sqrt{(x-a)^2 + (y-b)^2}$$

$$= \sqrt{(8-5)^2 + (8-4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25} = 5$$

$$\boxed{r = 5}$$